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Edition 2.0 2017-12

INTERNATIONAL STANDARD

Radiation protection instrumentation – Hand-held instruments for the detection and identification of radionuclides and for the estimation of ambient dose equivalent rate from photon radiation

IEC 62327:2017-12(en)

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RADIATION PROTECTION INSTRUMENTATION – HAND-HELD INSTRUMENTS FOR THE DETECTION AND IDENTIFICATION OF RADIONUCLIDES AND FOR THE ESTIMATION OF AMBIENT DOSE EQUIVALENT RATE FROM PHOTON RADIATION

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International Standard IEC 62327 has been prepared by subcommittee 45B: Radiation protection instrumentation, of IEC technical committee 45: Nuclear instrumentation.

This second edition cancels and replaces the first edition of IEC 62327, issued in 2006. It constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

a) addition of detailed methods of test;
b) revised identification test acceptance criteria for environmental tests;
c) changed format to match SC 45B template.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
45B/882/FDIS	45B/887/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

Illicit and inadvertent movement of radioactive materials in the form of radiation sources and contaminated metallurgical scrap has become a problem of increasing importance. Radioactive sources out of regulatory control, so-called “orphan sources”, have frequently caused serious radiation exposures and widespread contamination. Although illicit trafficking in nuclear and other radioactive materials is not a new phenomenon, concern about a nuclear “black market” has increased in the last few years particularly in view of its terrorist potential.

In response to the technical policy of the International Atomic Energy Agency (IAEA), the World Customs Organization (WCO) and the International Criminal Police Organization (Interpol) related to the detection and identification of special nuclear materials and security trends, nuclear instrumentation companies are developing and manufacturing radiation instrumentation to assist in the detection of illicit movement of radioactive and special nuclear materials. This type of instrumentation is widely used for security purposes at nuclear facilities, border control checkpoints, and international seaports and airports. However, to ensure that measurement results made at different locations are consistent, it is imperative that radiation instrumentation be designed to rigorous specifications based upon agreed performance requirements stated in this document. IEC standards have also been developed to address personal radiation detectors, radiation portal monitors, highly sensitive gamma and neutron detection systems, spectrometric personal radiation detectors, and backpack-based radiation detection and identification systems. Table 1 below contains a list of those standards.

Table 1 – IEC standards concerning instruments for the detection of illicit trafficking of radioactive material

Type of instrumentation	IEC number	Title of the standard
Body-worn	**62401**	Radiation protection instrumentation – Alarming Personal Radiation Devices (PRDs) for the detection of illicit trafficking of radioactive material
	62618	Radiation protection instrumentation – Spectroscopy-Based Alarming Personal Radiation Devices (SPRD) for detection of illicit trafficking of radioactive material
	62694	Radiation protection instrumentation – Backpack-type radiation detector (BRD) for detection of illicit trafficking of radioactive material
Portable or hand-held	**62327**	Radiation protection instrumentation – Hand-held instruments for the detection and identification of radionuclides and for the estimation of ambient dose equivalent rate from photon radiation
	62533	Radiation protection instrumentation – Highly sensitive hand-held instruments for photon detection of radioactive material
	62534	Radiation protection instrumentation – Highly sensitive hand-held instruments for neutron detection of radioactive material
Portal	**62244**	Radiation protection instrumentation – Installed radiation portal monitors (RPMs) for the detection of illicit trafficking of radioactive and nuclear materials
	62484	Radiation protection instrumentation – Spectroscopy-based portal monitors used for the detection and identification of illicit trafficking of radioactive material
Data format	**62755**	Radiation protection instrumentation – Data format for radiation instruments used in the detection of illicit trafficking of radioactive materials

RADIATION PROTECTION INSTRUMENTATION – HAND-HELD INSTRUMENTS FOR THE DETECTION AND IDENTIFICATION OF RADIONUCLIDES AND FOR THE ESTIMATION OF AMBIENT DOSE EQUIVALENT RATE FROM PHOTON RADIATION

1 Scope

This document applies to hand-held instruments used to detect and identify radionuclides and radioactive material, to estimate ambient dose equivalent rate from photon radiation, and optionally, to detect neutron radiation. They are commonly known as radionuclide identification devices or RIDs.

This document specifies general characteristics, general test procedures, radiation characteristics, as well as electrical, mechanical, safety, and environmental characteristics.

This document does not cover laboratory type, high-resolution photon spectrometers, or instruments covered by IEC 60846-1 (Portable workplace and environmental meters and monitors), IEC 60846-2 (photon dose (rate) meters) or IEC 61005 (neutron dose equivalent (rate) meters).

Table 8 provides a summary of requirements and relevant clauses.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-395:2014, *International Electrotechnical Vocabulary (IEV) – Part 395: Nuclear instrumentation: physical phenomena, basic concepts, instruments, systems, equipment and detectors*

IEC 60068-2-1, *Environmental testing – Part 2-1: Tests – Test A: Cold*

IEC 60068-2-2, *Environmental testing – Part 2-2: Tests – Test B: Dry heat*

IEC 60068-2-11, *Basic environmental testing procedures – Part 2 -11: Tests – Test Ka: Salt mist*

IEC 60068-2-14, *Environmental testing – Part 2-14: Tests – Test N: Change of temperature*

IEC 60068-2-18, *Environmental testing – Part 2-18: Tests – Test R and guidance: Water*

IEC 60068-2-27:2008, *Environmental testing – Part 2-27: Tests – Test Ea and guidance: Shock*

IEC 60068-2-64, *Environmental testing – Part 2-64: Tests – Test Fh: Vibration, broadband random and guidance*

IEC 60068-2-66, *Environmental testing – Part 2-66: Test methods – Test Cx: Damp heat, steady state (unsaturated pressurized vapour)*

IEC 60068-2-68, *Environmental testing – Part 2-68: Tests – Test L: Dust and sand*

IEC 60529, *Degrees of protection provided by enclosures (IP Code)*

IEC 60846-1, *Radiation protection instrumentation – Ambient and/or directional dose equivalent (rate) meters and/or monitors for beta, X and gamma radiation – Part 1: Portable workplace and environmental meters and monitors*

IEC 60846-2, *Radiation protection instrumentation – Ambient and/or directional dose equivalent (rate) meters and/or monitors for beta, X and gamma radiation – Part 2: High range beta and photon dose and dose rate portable instruments for emergency radiation protection purposes*

IEC 61000-4-2:2008, *Electromagnetic compatibility (EMC) – Part 4-2: Testing and measurement techniques – Electrostatic discharge immunity test*

IEC 61000-4-3:2006, *Electromagnetic compatibility (EMC) – Part 4-3: Testing and measurement techniques – Radiated, radio-frequency, electromagnetic field immunity test*
IEC 61000-4-3:2006/AMD1:2007
IEC 61000-4-3:2006/AMD2:2010

IEC 61000-4-6:2013, *Electromagnetic compatibility (EMC) – Part 4-6: Testing and measurement techniques – Immunity to conducted disturbances, induced by radio-frequency fields*

IEC 61000-4-8:2009, *Electromagnetic compatibility (EMC) – Part 4-8: Testing and measurement techniques – Power frequency magnetic field immunity test*

IEC 61005, *Radiation protection instrumentation – Neutron ambient dose equivalent (rate) meters*

IEC 61187, *Electrical and electronic measuring equipment – Documentation*

IEC 62706, *Radiation protection instrumentation – Environmental, electromagnetic and mechanical performance requirements*

IEC 62755, *Radiation protection instrumentation – Data format for radiation instruments used in the detection of illicit trafficking of radioactive materials*

3 Terms and definitions, abbreviated terms and symbols, quantities and units

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions, as well as those given in IEC 60050-395 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1.1
acceptance test
contractual test to prove to the customer that a device meets certain conditions of its specification

3.1.2
alarm
audible, visual, or other signal activated when the reading exceeds a pre-set value or falls outside a pre-set range

3.1.3
ambient dose equivalent (rate)
dose equivalent (rate) at a point in a radiation field, produced by the corresponding aligned and expanded field, in the ICRU sphere at a depth d, on the radius opposing the direction of the aligned field

Note 1 to entry: This definition does not include the notes that are part of definition IEC 60050-395:2014, 395-05-43.

3.1.4
background
radiation field in which there are no external sources present other than those in the natural radiation field at the location of the measurements

3.1.5
coefficient of variation
ratio of the standard deviation s to the arithmetic mean $\bar{x}$ of a set of n measurements x_i given by the following formula:

$$COV = \frac{s}{\bar{x}} = \frac{1}{\bar{x}}\sqrt{\frac{1}{n-1}\sum_{1}^{n}(x_i - \bar{x})^2}$$

3.1.6
confidence indicator
typically a numeric value assigned to a radionuclide identification result as to whether the identified radionuclide is present

3.1.7
conventionally true value
best estimate of the true value of a radiation field or source used for testing or calibration of equipment

3.1.8
error of indication
difference between the indicated value v of a quantity and the conventionally true value, CTV, of that quantity at the point of measurement

3.1.9
integration time
length of time used for collection of data to obtain a result

3.1.10
reference point
location marked on the instrument or described in the manual, used to establish radiation source-to-instrument distances and orientation for calibration or test purposes

3.1.11
relative intrinsic error
$\varepsilon_{\mathrm{REL}}$
difference between the instrument's reading, M, and the conventionally true value, CTV, of the quantity being measured divided by the conventionally true value when subjected to a specified reference quantity under specified reference conditions as given by the formula:

$$\varepsilon_{\mathrm{REL}} = \frac{M - CTV}{CTV}$$

3.1.12
restricted mode
operating mode used to access spectral data and to control the parameters that can affect the result of a measurement (for example: radionuclide library, routine function control, calibration parameters, alarm thresholds, etc.)

Note 1 to entry: May also be called expert mode.

Note 2 to entry: Access to this mode should be limited through password protection or other similar methods.

3.1.13
routine mode
operating mode that includes detection and identification of radionuclides, and estimation of the ambient dose equivalent rate

Note 1 to entry: May also be called easy mode.

3.1.14
stabilization time
duration, measured from the initial application of power, required for an instrument to indicate that it is operational

3.1.15
type test
conformity test on one or more specimens of a product representative of the production

3.1.16
uncertainty <of measurement>
parameter, associated with the result of a measurement, that characterizes the dispersion of the values that could reasonably be attributed to the measurand

Note 1 to entry: Uncertainty of measurement comprises, in general, many components. Some of these components may be evaluated from statistical distribution of the results of series of measurements and can be characterized by experimental standard deviations. The other components, which can also be characterized by experimental standard deviations, are evaluated from assumed probability distributions based on experience or other information.

Note 2 to entry: It is understood that the result of the measurement is the best estimate of the value of the measurand and that all components of uncertainty, including those arising from systematic effects, such as components associated with corrections and reference standards, contribute to the dispersion.

3.2 Abbreviated terms and symbols

COV	coefficient of variation
ESD	electrostatic discharge
DU	depleted uranium
HEU	highly enriched uranium
ICRU	International Commission on Radiation Units
lx	lux
NORM	naturally occurring radioactive material
RF	radio frequency
RGPu	reactor grade plutonium
RH	relative humidity
RID	radionuclide identification device
WGPu	weapons grade plutonium

3.3 Quantities and units

In this document, units of the International System (SI) are used[1]. The definitions of radiation quantities are given in IEC 60050-395.

The following units may also be used:

– for energy: electron-volt (symbol: eV), 1 eV = 1,602 x 10^{-19} J;
– for time: years (symbol: y), days (symbol: d), hours (symbol: h), minutes (symbol: min);
– for temperature: degrees Celsius (symbol: ºC), 0 ºC = 273,15 K.

Multiples and submultiples of SI units will be used, when practicable, according to the SI system.

4 General characteristics and requirements

4.1 General

RIDs are used for the detection, localization, and identification of radioactive material and for estimation of the ambient dose equivalent rate. They are hand-held, battery-powered instruments most commonly used for the detection and identification of illicit trafficking of radioactive material. They typically measure the photon energy spectrum and identify the radionuclide by comparison with an internal radionuclide library. RIDs may also detect neutrons.

The following are important design features:

– a display that is readable in low light (< 150 lx) and bright light (> 10 000 lx) conditions,
– user friendly controls, a menu structure that is simple and easy to follow, restricted access to critical operating parameters, and switches and other controls that minimize or prevent inadvertent operation.

NOTE During the development of this document, a desire for a RID to provide a simple user display for identification results was noted. The display discussed would provide identification results in the form of, for example, a green or red display to indicate that the radionuclide(s) identified was either not of interest or of possible interest, respectively. At the time of this revision, no specific functional requirements had been established. Requirements for the capability may be part of future revisions of this document.

4.2 Radiation detectors

RIDs may use multiple detectors to detect gamma radiation, identify radionuclides, and detect neutrons.

4.3 Personal protection alarm

4.3.1 Requirements

An alarm shall be provided to alert the user that the ambient gamma dose equivalent rate is above a threshold level. The alarm threshold shall be adjustable through the restricted mode. The alarm shall be both audible and visual with an "acknowledge" or other similar control to silence the audible function. It shall not be possible to deactivate both audible and visual indicators at the same time.

4.3.2 Method of test

a) Following the instructions provided by the manufacturer, set the personal protection alarm to activate at for example, 10 µSv h^{-1}.

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1 International Bureau of Weights and Measures: The International System of Units, 8th edition, 2006.

b) Expose the RID to a gamma source that provides an ambient dose equivalent rate reading 30 % above the alarm setting and verify the RID alarms.

c) While exposed to the radiation field, verify the alarm can be acknowledged. Access the RID menu to verify that it is not possible to disable both the audible and visual alarm indications simultaneously.

4.4 Stabilization time

4.4.1 Requirements

The manufacturer shall state the time required for the RID to become fully functional, i.e., to provide an indication of the ambient dose equivalent rate, be able to perform an identification, and detect neutrons, if applicable. The maximum time shall be less than 5 min from shutdown state or stand-by status (e.g., mechanically cooled detectors).

4.4.2 Method of test

Immediately after the manufacturer-stated stabilization time or within 5 min from switching the RID on either from shutdown state or stand-by status, expose the RID to ^{137}Cs producing an ambient dose equivalent rate of 0,5 µSv h^{-1} (± 30 %) above the background at the reference point of the RID and perform an identification. If the RID can detect neutrons, expose the RID to a neutron source moderated by high density polyethylene having a wall thickness of 4 cm.

The requirement is met if the RID is operational within 5 min of power-on as determined by providing an indication of the ambient dose equivalent rate, being able to perform an identification, and indicating the presence of neutrons, if neutron detection is provided.

4.5 Power supplies – battery

4.5.1 Requirements

The RID shall provide a visible indication of the battery condition, shall be able to detect and identify radionuclides, indicate ambient dose equivalent rate, and indicate the presence of neutrons (when applicable) for a minimum of 5 h under standard test conditions without indicating a low battery condition. The manufacturer shall also state the estimated operating time using the recommended battery(s) at standard test conditions as defined in Table 2 and at –20 °C.

NOTE When operated at low temperatures, the capacity of most types of batteries significantly decreases.

4.5.2 Method of test

Following the manual, verify that the manufacturer provided the expected continuous operating time using the recommended batteries at standard test temperature conditions and at –20 °C, and that the display includes a battery status indication.

Ensure the batteries are new or that rechargeable batteries are fully charged and after allowing the RID to stabilize in standard test conditions, perform a radionuclide identification using ^{137}Cs, and if applicable, expose the RID to a moderated neutron source as defined in 4.4.2. Record the results, including the indicated ambient dose equivalent rate with the source present and the neutron count rate, if provided. Repeat the process each hour for a total of 5 h. At the end of the 5 h period, perform another radionuclide identification and neutron response test, if applicable, with the same sources in the same position, and verify that the low battery indicator is not displayed.

The requirement is met when a low battery condition is not displayed and the RID is able to perform an identification with the same results, indicate the ambient dose equivalent rate with the source present within ± 30 % of the initial value, and if applicable, indicate the presence of neutrons. If a neutron count rate was provided, the count rate shall be within ± 30 % of the initial value.

4.6 Markings

All external controls, displays, and adjustments shall be identified according to their function. The following markings shall appear on the exterior of the RID or each major assembly (for example, detector probes) as appropriate:

- manufacturer and model number,
- serial number,
- location of the reference point(s), and
- function designation for controls, switches, and adjustments.

Markings shall be easily readable and permanently fixed under normal conditions of use including use of decontamination procedures (e.g., water and mild non-abrasive detergent).

4.7 Communication interface

4.7.1 Requirements

RIDs shall have the ability to transfer data to an external device, such as a computer. The transfer interface shall be fully described in the manual and should use a communications port that meets the requirements of Ethernet, USB, wireless, or other electronic means such as a removable media device. Consideration should be given to data security when using wireless data transfer techniques. The naming convention used for each file shall be defined in the manual.

4.7.2 Method of test

Following the manual, perform an identification and transfer the identification results to verify the RID can transfer data to an external device and that the transfer interface is described in the manual including the naming convention used for each file.

4.8 Data

4.8.1 Requirements

The RID shall have the ability to internally store at least 10 complete identification data files. Files produced as a data record for analysis should be in the format as specified by IEC 62755 and shall contain the following information as a minimum:

- RID manufacturer name, model, and serial number
- software and firmware version numbers
- gamma detector kind (e.g., NaI, LaBr, CZT, HPGe)
- date and time of measurement
- measured gamma-ray radiation level (e.g., ambient dose equivalent rate)
- background spectrum
- live time and real time for background spectrum
- measured spectrum
- live time and real time for measured spectrum
- energy calibration for the background and measured spectrum
- radionuclide identification results
- confidence indicator for radionuclide identification

If the RID has neutron detection capabilities, the data file shall include the following information:

- neutron detector kind (e.g., ^{3}He, Li-glass)

– measured neutron radiation levels (e.g., count rate).

4.8.2 Method of test

Verify the requirement by performing 10 radionuclide identifications. If the RID is equipped with a neutron detector, also expose the RID to a neutron source. When complete, transfer the stored results to a computer. On the computer, verify that each data file contains the information listed in the requirements and meets the data format requirements stated in IEC 62755.

5 General test procedures

5.1 Nature of test

Unless otherwise specified in the individual steps, all tests included in this document may be considered type tests. Certain tests may be considered acceptance tests by agreement between the customer and the manufacturer.

5.2 Statistical fluctuations

For the tests stated in Clauses 7, 8, and 9, the magnitude of the statistical fluctuations of the numerical indication arising from the random nature of radiation alone may be a significant fraction of the variation of the indication permitted in the test. The coefficient of variation (COV) for each mean reading should be less than or equal to 12 %.

The value of 12 % is chosen to help determine the results of a test when a series of readings and the associated mean are required for comparison purposes. Because RIDs tend to operate at low radiation levels meaning that testing should be done at similar radiation levels, it may be difficult to obtain COVs that are < 12 %. Processes such as increasing the applied radiation levels to an extent will typically reduce the COV to an acceptable level for the gamma detector, but the same may not occur for the neutron detector. The number of trials can also be increased, but that is an iterative process that may cause the number of readings to increase to an unreasonable amount and possibly hide the overall erratic nature of the RID under test.

The time interval between each reading shall be sufficient to ensure that the readings are independent. The time interval selected is dependent on the integration time of the RID and the update time of the display.

5.3 Standard test conditions

Except where otherwise specified, tests should be carried out under the standard test conditions shown in Table 2.

Table 2 – Standard test conditions

Influence quantity	Standard test conditions (unless otherwise indicated by the manufacturer)
Ambient temperature	18 °C to 25 °C
Relative humidity	≤ 75 %
Atmospheric pressure	70 kPa to 106,6 kPa
Gamma radiation background	Less than ambient dose equivalent rate of 0,15 μSv h^{-1}
Neutron radiation background	Less than 200 $s^{-1} \cdot m^{-2}$

5.4 Functionality test

5.4.1 General

The test methods and associated data collection requirements found in Clauses 7, 8 and 9 are stated in 5.4.2 through 5.4.4. They are used to verify the capability of a RID and not to estimate detection or identification probability and confidence in those estimates. Table 3 provides the techniques for analyzing test results.

5.4.2 Pre-test measurements

a) With the RID in position for test, expose it to a gamma and neutron radiation field using ^{133}Ba and ^{60}Co (or sources that emit low and high energy gamma rays) simultaneously, and a neutron source (when applicable). Source positions shall be marked or otherwise noted to ensure repeatability during test measurements.
b) Record a minimum of 10 readings (e.g., ambient dose equivalent rates, count rates) with the source(s) present. For RIDs with neutron detectors, if the count rate is not provided, the RID shall indicate the presence of neutrons.
c) Perform a series of 10 radionuclide identifications with the gamma sources present, and record the identification results including the confidence indicators. Collect at least one spectrum from the 10-trial series. Remove the sources.
d) Calculate and record the mean, standard deviation, and COV for the series of measurements. See 5.2, if needed, for additional information regarding the determination of the COV.
e) Establish the radiation response acceptance range (e.g., ambient dose equivalent rate, count rate) as ± 15 % of the calculated mean.

5.4.3 Intermediate measurements

a) While radioactive sources are not present, observe the RID and record any alarms, identifications or spurious indications.
b) To perform source testing, reposition each test source as needed per 5.4.2.
c) Record the same number of readings used for 5.4.2 (e.g., ambient dose equivalent rates, count rates, or alarm function) with the source(s) present.
d) Perform a series of 10 radionuclide identifications and record the identification results including the confidence indicators. Collect at least one spectrum from the 10-trial series. Remove the sources.
e) Calculate and record the mean, standard deviation, and COV for the measurement data.

5.4.4 Post-test measurements

a) Reposition each test source as needed per 5.4.2.
b) Record the same number of readings used for 5.4.2 (e.g., ambient dose equivalent rates, count rates, alarm function) with the source(s) present.
c) Perform a series of 10 radionuclide identifications and record the identification results including the confidence indicators. Collect at least one spectrum from the 10-trial series. Remove the sources.
d) Calculate and record the mean, standard deviation, and COV for the measurement data.

Table 3 – Test result analysis

Verification test	Analysis technique
Functionality	The results are acceptable if there are no unexpected alarms or fault indications during the test
Identification results	The identification results are acceptable if there are complete and correct results in 9 out of 10 trials at each test point.
Radiation response	The mean response at each test point shall be within the acceptance range of ± 15 % and the COV should be less than 12 %.

6 Radiation detection requirements

6.1 Ambient dose equivalent rate

6.1.1 Requirements

Under standard test conditions, the relative intrinsic error of the response of the RID to the photon radiation from ^{241}Am, ^{137}Cs, and ^{60}Co shall not be greater than ± 30 % for all ambient dose equivalent rates from 1 µSv h^{-1} up to the manufacturer-stated maximum ambient dose equivalent rate. If the manufacturer states that the RID can be used for photon radiation protection purposes, it shall meet the requirements stated in IEC 60846-1 or IEC 60846-2.

6.1.2 Method of test

Expose the RID to ambient dose equivalent rates of 1 µSv h^{-1}, 10 µSv h^{-1}, and 70 % of the manufacturer-stated maximum ambient dose equivalent rate range from each of the required radionuclides at the reference point and verify that the readings are within ± 30 % of the applied ambient dose equivalent rate.

6.2 Gamma source localization

6.2.1 Requirements

While attempting to localize (or search for) a radiation source, RIDs shall provide an indication when the radiation field changes. Indication should include audible tones, visible display using a moving chart, or changes to a numerical value.

6.2.2 Method of test

The requirement is verified using ^{241}Am, ^{137}Cs, and ^{60}Co moving at a speed of 0,5 m·s^{-1} (± 10 %) producing a radiation field of 0,5 µSv h^{-1} (± 20 %) above background at the reference point of the RID at the distance of closest approach to the source (or test point) of 1 m.

The distance of closest approach may vary between 1 m and 2 m to allow for variability in source activity. If the distance of closest approach, d (expressed in m), is not 1 m the passage speed v (expressed in m·s^{-1}) shall be adjusted using:

$$v = v_0 \times d/d_0$$

where v_0 = 0,5 m·s^{-1} and d_0 = 1 m.

During testing, the source or RID shall be moved from a position where the source is not detectable, through the test point, to a position where the source is again not detectable. A controlled linear motion system is recommended. The RID's orientation with respect to the source for this test shall be the same as the orientation when performing a scan and shall be described in the test record.

a) Establish the source-to-detector distance that provides an ambient dose equivalent rate of 0,5 μSv h^{-1} (± 20 %) above background at the reference point of the RID using ^{241}Am, ^{137}Cs, and ^{60}Co.

The following steps are based on the source moving with the RID stationary.

b) Using the ^{137}Cs source, when the RID is ready and operating as it would for search or scan, move the ^{137}Cs source through the test point at a speed of 0,5 m·s^{-1} ± 10 % or at the determined speed.
c) As the source approaches the test point, continuously observe the RID's response and record whether it provides an indication(s) of the increased radiation field. It is acceptable for an alarm or indication to occur as the source approaches the RID. The indication or alarm shall activate no later than 1 s after the source passes through the test point (point of closest approach to the reference point).
d) Acknowledge the alarm (if applicable) and repeat the process nine additional times with a minimum of 10 s between each trial. Acceptable results are when an indication or alarm is activated as described in 6.2.1 in at least 9 out of 10 trials.

   NOTE This requirement is meant to demonstrate the capability of the RID, not to estimate detection probability and confidence in those estimates.

e) Repeat the entire process using ^{241}Am and ^{60}Co.

6.3 Over-range characteristics for ambient dose equivalent rate

6.3.1 Requirements

RIDs shall indicate that an over-range condition exists when the ambient dose equivalent rate is greater than the manufacturer-stated maximum ambient dose equivalent rate. The indicators for over range conditions must be clearly visible and/or audible, and be different than that provided for gamma source localization as required in 6.2.

6.3.2 Method of test

a) Using ^{137}Cs, perform a single identification in accordance with the method of test found in 6.6.3.2.
b) Using ^{137}Cs, expose the RID to an ambient dose equivalent rate that is 1,5 times that of the manufacturer-stated maximum ambient dose equivalent rate for 5 min.
c) The RID shall indicate that an over-range condition exists within 5 s of the step change and shall remain in that condition for the entire 5 min exposure period.
d) After the 5-min exposure, reduce the radiation field to the pre-test value. The RID shall operate normally within 5 min.
e) To verify that no changes took place as a result of the test, repeat a) and verify the RID identifies the radionuclide used in the initial test.

6.4 Neutron detection

6.4.1 Requirements

RIDs that have neutron detection capabilities shall alarm or provide an indication when the neutron radiation field changes from ambient to that with a neutron source present within a time of 2 s after the RID passes through the test point. The neutron alarm or indicator shall be different than the personal protection alarm.

This requirement is verified using a ^{252}Cf source having an emission rate of 2×10^4 (± 20 %) neutron s^{-1} that is surrounded by a spherical high-density polyethylene (0,93 to 0,97 g/cm^3) moderator having a wall thickness of 4 cm and an inner cavity diameter not larger than 3 cm.

The point of closest approach of the source assembly shall be 25 cm from centre of the neutron source placed in the centre of the sphere to the reference point of the neutron detector. This is the test point.

If the manufacturer states that the RID can be used for neutron radiation protection purposes, it shall meet the requirements stated in IEC 61005.

6.4.2 Method of test

The neutron source assembly or RID shall be moved from a position where the source is not detectable, through the test point, to a position where the source is again not detectable. A controlled linear motion system is recommended.

The RID's orientation with respect to the source assembly for this test shall be the same as the orientation when performing a scan and shall be described in the test record.

The source assembly and the RID shall be at least 1 m from the ground or floor and 2 m from any other item or surface (e.g., storage cabinets, walls) to reduce scatter. Support material such as the source/source assembly holder shall not make use of hydrogenous material (e.g., foam, plastic). Hand phantoms used to simulate a person holding the RID are permitted and should be provided by the manufacturer.

The following steps are based on the neutron source assembly defined in 6.4.1 moving adjacent to the RID through the test point.

a) When the RID is ready and operating as it would for search or scan, move the neutron source assembly through the test point at a speed of 0,5 $m \cdot s^{-1}$ ± 10 %.
b) As the source assembly approaches the test point, continuously observe the RID's response and record whether it provides an indication(s) of the neutron source. It is acceptable for the neutron alarm or indication to occur as the source assembly approaches the RID.
c) Verify whether the neutron alarm or indication occurs within 2 s of the neutron source assembly passing through the test point.
d) Repeat the process nine additional times with a minimum of 10 s between each trial.
e) Acceptable results are when the neutron indication is activated in at least 9 out of 10 trials.

NOTE This requirement is meant to demonstrate the capability of the RID, not to estimate detection probability and confidence in those estimates.

6.5 Neutron indication in the presence of photons

6.5.1 Requirements

An ambient gamma dose equivalent rate of 0,1 mSv h^{-1} above background shall not cause a RID with neutron detection capabilities to indicate the presence of neutrons or activate a neutron alarm. In addition, the RID shall be able to detect the presence of neutrons or activate a neutron alarm while being exposed to the 0,1 mSv h^{-1} gamma radiation field whether or not the RID is in gamma over-range condition.

6.5.2 Method of test

a) Expose the RID to photons from ^{137}Cs providing an ambient dose equivalent rate of 0,1 mSv h^{-1} (± 20 %) at the neutron reference point as defined by the manufacturer.

   NOTE 0,1 mSv h^{-1} was selected based on typical ambient dose equivalent rates produced by commonly used medical radionuclides that may be seen during use. ^{137}Cs was selected due to its photon energy being close to the maximum photon energy emission from some of the commonly used medical radionuclides.

b) Verify that neutrons are not indicated as a result of the photons during a continuous exposure time of 1 min. In order to eliminate dependence on the neutron detector geometry, the distance between the ^{137}Cs source and the reference position should be at least 50 cm.
c) While the RID is exposed to the ^{137}Cs source, expose the RID to the neutron source using the same technique used in 6.4.2 without shielding the ^{137}Cs source.

d) The presence of neutrons shall be indicated with the ^{137}Cs source present.

e) Repeat the process for a total of 10 trials.

f) The results are acceptable when the requirements are met in at least 9 out of 10 trials.

6.6 Radionuclide identification

6.6.1 Radionuclide identification library

6.6.1.1 Requirements

The manufacturer shall provide a list of the radionuclides that are in the library. The list of radionuclides shall include those shown in Table 4, as a minimum.

6.6.1.2 Method of test

Verify that the requirement is met by review of manufacturers' provided information and direct observation of the RID library.

Table 4 – Radionuclide library

Radionuclide (indicates test source materials)	
^{241}Am	^{40}K (KCl or KOH)
^{133}Ba	^{201}Tl
^{60}Co	^{226}Ra
^{137}Cs	^{232}Th
^{67}Ga	^{238}U (DU)
^{131}I	^{235}U (HEU)
^{99m}Tc	^{239}Pu (WGPu)

NOTE For this document, the composition of materials is: DU < 0,4 % ^{235}U, HEU ≥ 90 % ^{235}U, and WGPu ≤ 6,5 % ^{240}Pu and > 93 % ^{239}Pu.

6.6.2 Identification results

6.6.2.1 Requirements

A RID shall display the result of a radionuclide identification (e.g., "Cs-137"). When identifying different uranium isotopes, it is acceptable to display "Uranium". It is also acceptable to display "Plutonium" for RGPu and WGPu identification results.

The RID shall display for example, "not identified" or "unknown", if the RID indicates the presence of a source, but is not able to identify a radionuclide.

An indication may be displayed as to the confidence of the radionuclides identified. The manufacturer shall explain the meaning of the confidence indicator.

The RID shall indicate when the radiation field is too high for radionuclide identification.

6.6.2.2 Method of test

a) Perform a single identification using ^{137}Cs at an ambient dose equivalent rate of 0,5 µSv h^{-1} above background at the reference point within an integration time of 1 min or that stated by the manufacturer, whichever is shorter, and record the results to verify the identification requirement.

b) Expose the RID to a radionuclide that is not included in the library and verify that it provides an indication such as "not identified", or "unknown". The radionuclide should have at least one gamma line that is within the energy range of the RID. A radionuclide (agreed on with the manufacturer) may be temporarily removed from the library to perform this verification test.

c) If a confidence indicator is provided, perform a radionuclide identification to verify that it is displayed. Review the manual to verify that the manufacturer provided an explanation of the confidence indicator.

d) Repeat item a) increasing the ambient dose equivalent rate until the RID indicates that the rate is too high for identification.

   This is an iterative process to determine whether the RID provides the required indication and could require several identifications. This is not a test to verify the correctness of the RID's identification capabilities.

6.6.3 Radionuclide and radioactive material identification

6.6.3.1 Requirements

A RID shall be able to identify the items listed in Table 4 at an ambient dose equivalent rate of 0,5 µSv h^{-1} above background measured at the gamma reference point of the RID within an integration time of 1 min or that stated by the manufacturer, whichever is shorter.

Table 5 and Table 6 provide guidance regarding identification performance and list the daughters and possible impurities, respectively. Following Table 5, identification results shall be reported as either a) complete and correct, b) incomplete, c) incorrect or, d) incomplete and incorrect.

The medical radionuclides (^{99m}Tc, ^{131}I, and ^{201}Tl) shall be surrounded by 8 cm of polymethyl methacrylate (PMMA) for test purposes to represent in-vivo conditions.

NOTE 1 Actual activities for medical radionuclides given to patients are much greater than those required in this document.

NOTE 2 Identification requirements are based on the sources and source configurations defined in this document. When used operationally, the configuration of a source or object being analysed is typically unknown which may cause other radionuclides or isotopes to be identified.

6.6.3.2 Method of test

Expose the RID to each item listed in Table 4. The ambient dose equivalent rate at the reference point of the RID, obtained with an independent radiation measuring device such as an ambient dose equivalent meter, shall be 0,5 µSv h^{-1} (± 30 %) above background.

The test shall consist of 10 trials for each radionuclide. The performance is acceptable when the identification results are complete and correct in at least 9 of 10 trials.

NOTE The test methods and associated data collection requirements are not meant to estimate detection or identification probability and confidence in those estimates.

Test scheduling needs to account for the short half-life of medical radionuclides.

Table 5 – Guidance regarding identification performance

Complete and correct	Radionuclide "X" identified as (→) "X" Radionuclides "X + Y" identified as "X + Y" For example: - $^{235}U \rightarrow {}^{235}U$ - $^{235}U \rightarrow {}^{235}U + {}^{40}K$ - $^{235}U \rightarrow {}^{235}U + {}^{40}K + {}^{232}Th$ - $^{235}U \rightarrow {}^{235}U + {}^{40}K + {}^{232}Th + {}^{226}Ra$ - $^{235}U + {}^{67}Ga \rightarrow {}^{235}U + {}^{67}Ga + {}^{40}K + {}^{232}Th + {}^{226}Ra$ Complete and correct may also include daughter(s) and impurities of the target radionuclide(s). Table 6 provides a list of daughters and possible impurities.
Incomplete	Radionuclides "X + Y" identified as "X" or "Y" For example: - $^{235}U + {}^{226}Ra \rightarrow {}^{226}Ra$
Incorrect	Radionuclide "X" identified as "X + Y" For example: - $^{235}U \rightarrow {}^{235}U + {}^{237}Np$ - $^{67}Ga \rightarrow {}^{235}U + {}^{67}Ga$
Incomplete and incorrect	Radionuclide "X" being identified as "Z" Radionuclides "X + Y" identified as "Z + A" For example: - $^{235}U \rightarrow {}^{67}Ga$ - $^{235}U + {}^{137}Cs \rightarrow {}^{99m}Tc + {}^{133}Ba$

Table 6 – List of likely daughters and possible impurities

Radionuclide(s)/materials	**Daughters and possible impurities**
^{201}Tl	^{202}Tl
DU (^{238}U)	^{235}U, ^{226}Ra
WGPu (^{239}Pu)	^{242}Pu, ^{241}Pu, ^{240}Pu, ^{238}Pu, ^{241}Am, ^{237}U, ^{233}U
HEU (^{235}U)	^{238}U, ^{234m}Pa
^{99m}Tc	^{99}Mo
^{232}Th	^{228}Th, ^{232}U
^{226}Ra	^{214}Bi, ^{214}Pb

6.6.4 Identification of mixed radioactive materials

6.6.4.1 Requirements

The following combined radionuclides shall be identified within 1 min or as specified by the manufacturer, whichever is shorter. Each radionuclide shall produce an ambient dose equivalent rate (unshielded) at the reference point of 0,5 µSv h^{-1} (± 30 %) above background:

– ^{131}I + WGPu
– NORM (or surrogate)
– NORM + HEU
– NORM + WGPu

6.6.4.2 Method of test

Expose the RID to each combination described in 6.6.4.1. The test process is defined in 6.6.3.2.

Except when testing with NORM alone, each source within each combination shall produce 0,5 µSv h^{-1} (±30 %) above background at the reference point of the RID. The performance is acceptable when the RID identifies HEU or WGPu as shown in Table 6 and as appropriate based on the combination tested in at least 9 of 10 trials.

For NORM, bulk NORM or preferably a NORM surrogate made from ^{226}Ra and ^{232}Th point sources shielded by 9 cm of PMMA, should be used. KCl (or KOH) may be added when using the surrogate NORM to better represent NORM found in commerce. If used, the total ambient dose equivalent rate from the other sources will need to be reduced to maintain the rate at 0,5 µSv h^{-1} (± 30 %).

When testing with NORM alone, the total ambient dose equivalent rate at the reference point of the RID for either bulk NORM or a NORM surrogate shall be 0,5 µSv h^{-1} (± 30 %) above background. The identification test shall be performed using the same process defined in 6.6.3.2 and the performance is acceptable when the identification results do not indicate the presence of radionuclides that are not present in at least 9 of 10 trials.

7 Environmental requirements

7.1 General

RIDs shall comply with IEC 62706 concerning the ambient temperature, relative humidity, and other metrological requirements for hand-held instrumentation. To ensure that each requirement is met, ^{133}Ba and ^{60}Co are used for the gamma detector and a neutron source (e.g., ^{252}Cf, AmBe) is used for the neutron detector.

7.2 Ambient temperature

7.2.1 Requirements

Over the temperature range specified in IEC 62706 for hand-held devices (–20 °C to +50 °C), the RID shall function correctly including performing identifications, responding to neutron sources (when applicable), and maintaining an acceptable indication of ambient dose equivalent rate. There shall be no visible external damage to the RID and all control functions shall be verified to be operating correctly.

7.2.2 Method of test

Set up the temperature test system as stated in IEC 60068-2-1 and IEC 60068-2-2, cold and dry heat, respectively. Follow the ambient temperature method of test specified in IEC 62706.

For this test, an external power supply may be used to provide power to the RID ensuring that it remains on for the duration of the test.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2 at 20 °C ± 2 °C (nominal temperature). During the test, perform the intermediate measurements as described in 5.4.3 at –10 °C, 0 °C, 10 °C, 30 °C, and 40 °C after a 2 h soak at each temperature, as well as at the low and high temperature extremes at the beginning, middle, and end of the 16 h soak. The temperature test shall be performed from the low to the high extremes, then back to the nominal temperature.

Following return to the nominal temperature, perform the post-test measurements as described in 5.4.4 and analyse the results in accordance with Table 3.

7.3 Temperature shock

7.3.1 Requirements

Over the temperature range specified in IEC 62706 for hand-held devices (–20 °C to +50 °C), the RID shall function correctly including performing identifications, responding to neutron sources (when applicable), and maintaining an acceptable indication of ambient dose equivalent rate within 1 h of exposure to rapid temperature changes from 20 °C to –20 °C, –20 °C to 20 °C, 20 °C to 50 °C, and 50 °C to 20 °C with each change being made in less than 5 min. There shall be no visible external damage to the RID and all control functions shall be verified to be operating correctly.

7.3.2 Method of test

For this test, an external power supply may be used to provide power to the RID ensuring that it remains on for the duration of the test.

Set up the temperature test system as stated in IEC 60068-2-14, change of temperature, and follow the temperature shock method of test specified in IEC 62706.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2 at 20 °C ± 2 °C (nominal temperature). During the test, perform the intermediate measurements (3 identifications only) as described in 5.4.3 at 5 min, 15 min, 30 min, and 1 h after each temperature change takes place. The temperature test shall be performed to and from the low temperature set point first.

Following return to the nominal temperature, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3.

7.4 Relative humidity

7.4.1 Requirements

Over the range of relative humidity specified in IEC 62706 for hand-held devices (40 % to 93 % at 35 °C), the RID shall function correctly including performing identifications, responding to neutron sources (when applicable), and maintaining an acceptable indication of ambient dose equivalent rate. As a result of the humidity change, the indication of the magnitude of the radiation field or of the ambient dose equivalent rate (if provided) shall not change as described in the method of test.

There shall be no visible external damage to the RID, and all control functions shall be verified to be operating correctly.

7.4.2 Method of test

For this test, an external power supply may be used to provide power to the RID ensuring that it remains on for the duration of the test.

Set up the temperature/humidity test system as stated in IEC 60068-2-66, and follow the relative humidity method of test specified in IEC 62706.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2 at 20 °C ± 2 °C and RH of < 75 % (nominal temperature and humidity). During the test, perform the intermediate measurements as described in 5.4.3 at beginning, middle and end of the 16 h high humidity soak period.

Following return to the nominal temperature and humidity, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3.

7.5 Low/high temperature start-up

7.5.1 Requirements

RIDs shall be able to operate when switched on at the cold temperature limit (−20 °C) or the high temperature limit (50 °C).

7.5.2 Method of test

For this test, an external power supply shall not be used.

Set up the temperature test system as stated in IEC 60068-2-1 and IEC 60068-2-2, cold and dry heat, respectively, and follow the low/high temperature start-up method of test specified in IEC 62706.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. During the test, perform the intermediate measurements as described in 5.4.3 after a period of 4 h at each temperature test point.

Following the final return to 20 °C, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3.

7.6 Moisture and dust protection

7.6.1 Requirements

The RID case design shall meet the IP classification requirements stated in IEC 62706 for hand-held devices (IP 53).

7.6.2 Method of test – dust

Set up the dust system and RID as the device under test as defined in IEC 60068-2-68. The test shall be performed based on test protocol La2 for a period of 1 h.

To verify the performance of the RID against the requirement, perform the pre-test measurements as described in 5.4.2. Following exposure to the dust environment, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3. For this test, also inspect the unit to determine the extent of dust ingress. Attention shall be paid to the battery compartment, display, and any other easily accessed portions of the RID.

7.6.3 Method of test – moisture

Set up the moisture system and RID as the device under test as defined in IEC 60529. Additional information is available in IEC 60068-2-18. All sides of the RID shall be exposed to the water spray starting with the nozzle directly pointed to the centre of the side being exposed. The spray nozzle or RID shall be moved during the test such that the water direction is changed up to +60° and −60° in two orthogonal orientations relative to each side of the RID. The test duration shall be 1 min per m^2 calculated based on the surface area of the RID with a minimum duration of 5 min.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. Observe the response of the RID while being exposed to the water spray and record any functional changes that may occur (e.g., alarms, fault indications). Following exposure to the moisture environment, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3. For this test, also inspect the unit to determine the extent of moisture ingress. Attention shall be paid to the battery compartment, display, and any other easily accessed portions of the RID.

For testing against salt water spray, the RIDs shall exhibit the same performance after being subjected to a salt water spray as defined in IEC 60068-2-11, using a conditioning duration of 16 h. The RID may not be powered during the test.

8 Mechanical requirements

8.1 General

RIDs shall comply with IEC 62706 concerning the mechanical requirements for hand-held or non-installed instrumentation.

8.2 Vibration

8.2.1 Requirements

RIDs shall withstand exposure to vibrations associated with the operation of hand-held equipment as defined in IEC 62706, vibration test, hand-held, body worn, portable, and transportable requirements.

8.2.2 Method of test

Set up the vibration system and RID as the device under test as defined in IEC 60068-2-64. The test specifics are 0,01 g^2 Hz^{-1} (spectral density) using 5 Hz and 500 Hz as the frequency endpoints for a period of 1 h.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. Observe the response of the RID while being exposed to the vibration and record any functional changes that may occur (e.g., alarms, fault indications). Following vibration, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3. In addition, inspect the RID to ensure that the physical condition was not affected by vibration (e.g., solder joints shall hold; nuts and bolts shall not come loose).

8.3 Mechanical shock

8.3.1 Requirements

RIDs shall withstand exposure to shock pulses in accordance with the mechanical shock requirements defined in IEC 62706.

The physical condition of a RID shall not be affected by these shocks (e.g., solder joints shall hold; nuts and bolts shall not come loose).

8.3.2 Method of test

Set up the mechanical shock system and RID as the device under test as defined in IEC 60068-2-27. The test specifics are 50 g peak acceleration for a nominal 11 ms in each of three mutually orthogonal axes.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. After each test orientation, perform the intermediate measurements as described in 5.4.3. Following the test, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3. In addition, inspect the RID to ensure that the physical condition was not affected by the shocks (e.g., solder joints shall hold; nuts and bolts shall not come loose).

8.4 Impact (microphonics)

8.4.1 Requirements

RIDs shall be unaffected by microphonic conditions such as those that may occur from low intensity impacts from sharp contact with hard surfaces in accordance with the microphonic/impact requirements for hand-held and body worn devices defined in IEC 62706.

8.4.2 Method of test

Follow the method of test defined in IEC 62706.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. Observe the response of the RID while being exposed to the impacts and record any functional changes that may occur (e.g., alarms, fault indications). Following the test, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3.

9 Electromagnetic requirements

9.1 General

RIDs shall comply with IEC 62706 concerning the radio frequency susceptibility and other electromagnetic requirements for hand-held instrumentation.

9.2 Electrostatic Discharge (ESD)

9.2.1 Requirements

RIDs shall not be affected by exposure to electrostatic discharges at intensities of up to 6 kV using the contact discharge technique in accordance with the electrostatic requirements defined in IEC 62706.

9.2.2 Method of test

Set up the test conditions in accordance with IEC 61000-4-2 and using the method of test defined in IEC 62706 for battery powered instruments.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. Without the radioactive sources present, observe the response of the RID while being exposed to the discharges and record any functional changes that may occur (e.g., alarms, fault indications). Following the ESD exposure, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3.

9.3 Radio Frequency (RF)

9.3.1 Requirements

RIDs should not be affected by RF fields in accordance with the radiofrequency immunity requirements for all other instrument types as defined in IEC 62706 with the following changes:

– Frequency range from 80 MHz to 2 500 MHz at field intensities of 10 $V \cdot m^{-1}$ from 80 MHz to 1 000 MHz and 3 $V \cdot m^{-1}$ from 1 000 MHz to 2 500 MHz.

NOTE Wireless interface technologies may not work in the presence of RF.

9.3.2 Method of test

Set up the test conditions in accordance with IEC 61000-4-3 and use the method of test defined in IEC 62706. The RID should be oriented with the display facing the RF emission source.

The test orientation is based on where a RID may be most likely susceptible to RF. Other orientations may be selected based on a physical inspection of the RID case, e.g., cable connections, switch locations.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. Without moving the radiation test sources, observe the response of the RID while being exposed to the radio frequency conditions and record any functional changes that may occur (e.g., alarms, display changes, fault indications). Remove the radiation test sources and repeat the radiofrequency exposure while observing the response of the RID. Record any functional changes that may occur (e.g., alarms, display changes, fault indications).

Following the completion of the test, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3.

9.4 Radiated RF emissions

9.4.1 Requirements

The electromagnetic fields emitted by the RID at 3 m shall be less than what are shown in Table 7 in accordance with IEC 62706.

Table 7 – Emission frequency limits

Frequency MHz	**Field strength** $\mu V \cdot m^{-1}$
30 to 88	100
88 to 216	150
216 to 960	200
> 960	500

9.4.2 Method of test

a) Place the RID in an area with a low and controlled radio frequency environment.
b) Position an antenna 3 m from the RID in an orientation where RF emissions may be likely (e.g., openings, cable penetrations, display). A distance of 1 m may be used with the field strength levels adjusted accordingly.

   The device under evaluation may be rotated using a turntable while monitoring the RF spectrum to identify the location of RF emissions.
c) With the RID turned off, collect a background RF spectrum using a scanning bandwidth of 120 kHz over a range from 30 MHz to 1 000 MHz.
d) Once the RID is operational, perform an RF scan with the antenna in the vertical orientation.
e) Record emissions that are greater than the values stated in Table 7.
f) If no RF emissions are observed, proceed to step g). If observed RF emissions are greater than the values stated in Table 7, the RID fails the test and no additional testing is required.
g) Rotate the antenna to the horizontal orientation and repeat steps d) and e).

9.5 Conducted disturbances

9.5.1 Requirements

RIDs shall not be affected by RF fields that can be conducted onto the instrument through an external conducting cable in accordance with IEC 62706, immunity from conducted RF. RIDs that do not have at least one external conducting cable are excluded.

9.5.2 Method of test

Set up the test conditions in accordance with IEC 61000-4-6 and use the method of test defined in IEC 62706.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. Without moving the radiation test sources, observe the response of the RID while being exposed to the radio frequency conditions and record any functional changes that may occur (e.g., alarms, display changes, fault indications). Remove the radiation test sources and repeat the radiofrequency exposure while observing the response of the RID. Record any functional changes that may occur (e.g., alarms, display changes, fault indications).

Following the completion of the test, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3.

9.6 Magnetic fields

9.6.1 Requirements

The RID shall be fully functional when exposed to a 100 $A{\cdot}m^{-1}$ 50 Hz or 60 Hz magnetic field in accordance with IEC 62706.

9.6.2 Method of test

Set up the test conditions in accordance with IEC 61000-4-8 and use the method of test defined in IEC 62706.

To verify the performance of the RID against the requirements, perform the pre-test measurements as described in 5.4.2. Without moving the radiation test sources, observe the response of the RID while being exposed to the magnetic field and record any functional changes that may occur (e.g., alarms, fault indications). While being exposed to the magnetic field, perform the intermediate measurements as described in 5.4.3 for each test orientation.

Remove the radiation test sources; repeat the magnetic field exposure while observing the response of the RID in each orientation. Record any functional changes that may occur (e.g., alarms, fault indications).

Following the completion of the test, perform the post-test measurements as described in 5.4.4, and analyse the results in accordance with Table 3.

10 Documentation

10.1 Operation and maintenance manual

Each RID shall be supplied with an appropriate instruction manual in accordance with IEC 61187. The instructions for use of the RID shall contain a description of the construction, function, operation and manipulation of the RID. The manual shall include the following information, as a minimum:

a) Contact information for the manufacturer including name, address, telephone number, fax number, e-mail address, etc.
b) Model name or number
c) Type of radiation detector(s) used (e.g., NaI, CZT, HPGe, ^{3}He). If pressurized detectors are used (e.g., ^{3}He), the manufacturer shall state the dimensions, pressure, and active volume of the detector
d) Size and weight (with batteries installed)

e) Manufacturer’s website
f) Battery requirements
g) Recommended operational parameters such as: false alarm probability, alarm thresholds, operating parameters, stabilisation time, and radionuclide library(s)
h) Static identification measurement time
i) General description including functional controls of the RID
j) Case specification classification (i.e., IP classification)
k) Inclusion of any hazardous material that may be subject to regulation (such as radionuclide check source, pressurized gases, corrosive materials)
l) List of radionuclides that are identified by the RID
m) Description of the confidence indicator
n) Ambient dose equivalent rate range
o) Ambient dose equivalent rate calibration results
p) Over-range ambient dose equivalent rate value
q) Operating temperature range
r) Method and type of information transmitted from the RID to a computer.

10.2 Test certificate

A certificate shall accompany each RID providing the following information:

a) Manufacturer’s name or registered trademark
b) Contact information for the manufacturer including address, telephone number, fax number, e-mail address, etc.
c) Model name or number, serial number, and software and firmware version numbers
d) List of radionuclides to which the RID was tested
e) Ambient dose equivalent rate range to which the RID was tested.

10.3 Declaration of conformity

A declaration of conformity to each applicable clause of this document (and all other standards to which the RID may comply) should be provided. The declaration shall include the name and address of the manufacturer and its representative, the equipment type, and serial number.

Table 8 – Summary of performance requirements

Parameter	Performance requirement or specification	Relevant sub-clause
General	RIDs shall incorporate a display that is readable in low light and bright light conditions, and user-friendly controls including a menu structure that is simple and easy to follow, restricted access to critical operating parameters, and switches and other controls that minimize or prevent inadvertent operation.	4.1
Radiation detectors	RIDs may use multiple detectors to detect gamma radiation, identify radionuclides, and detect neutrons.	4.2
Personal protection alarm	An alarm shall be provided to alert the user that ambient gamma dose equivalent rates are above a threshold level.	4.3
Stabilization time	The manufacturer shall state the time required for the RID to become fully functional.	4.4
Power supplies – battery	RIDs shall be fully functional for a minimum of 5 h under standard test conditions.	4.5
Markings	All external RID controls, displays, and adjustments shall be identified according to their function.	4.6
Communication interface	RIDs shall have the ability to transfer data to an external device, such as a computer.	4.7
Data	RIDs shall have the ability to internally store at least 10 complete identification data files. Data files shall be in the format as specified by IEC 62755.	4.8
Ambient dose equivalent rate	Under standard test conditions, the relative intrinsic error of the response of the RID to the photon radiation from ^{241}Am, ^{137}Cs, and ^{60}Co shall not be greater than ± 30 % for all ambient dose equivalent rates from 1 µSv h^{-1} up to the manufacturer-stated maximum ambient dose equivalent rate. If the manufacturer states that the RID can be used for photon radiation protection purposes, it shall meet the requirements stated in IEC 60846-1 or IEC 60846-2.	6.1
Source localization	While attempting to localize (or search for) a radiation source, RIDs shall provide an indication when the radiation field changes. Indication should include audible tones, visible display using a moving chart, or changes to a numerical value.	6.2
Over-range characteristics for ambient dose equivalent rate	RIDs shall indicate that an over-range condition exists when the ambient dose equivalent rate is greater than the manufacturer-stated maximum ambient dose equivalent rate. The indicators for over range conditions must be clearly visible and/or audible, and be different than that provided for gamma source localization as required in 6.2.	6.3
Neutron detection	RIDs that have neutron detection capabilities shall alarm or provide an indication when the neutron radiation field changes from ambient to that with a neutron source present within a time of 2 s after the RID passes through the test point. The neutron alarm or indicator shall be different than the personal protection alarm.	6.4
Neutron indication in the presence of photons	An ambient gamma dose equivalent rate of 0,1 mSv h^{-1} above background shall not cause a RID with neutron detection capabilities to indicate the presence of neutrons or activate a neutron alarm. In addition, the RID shall be able to detect the presence of neutrons or activate a neutron alarm while being exposed to the 0,1 mSv h^{-1} gamma radiation field whether or not the RID is in gamma over-range condition.	6.5
Radionuclide identification library	The manufacturer shall provide a list of the radionuclides that are in the library. The list of radionuclides and materials shall include those shown in Table 4, as a minimum.	6.6.1
Identification results	A RID shall display the result of a radionuclide identification (e.g., “Cs-137”). When identifying different uranium isotopes, it is acceptable to display “Uranium”. It is also acceptable to display “Plutonium” for RGPu and WGPu identification results. The RID shall display for example, “not identified” or “unknown” if the RID indicates the presence of a source but is not able to identify a radionuclide. An indication may be displayed as to the confidence of the radionuclides identified. The manufacturer shall explain the meaning of the confidence indicator. The RID shall indicate when the radiation field is too high for radionuclide identification.	6.6.2

Parameter	Performance requirement or specification	Relevant sub-clause
Radionuclide and radioactive material identification	A RID shall be able to identify the items listed in Table 4 at an ambient dose equivalent rate of 0,5 µSv h^{-1} above background measured at the gamma reference point of the RID within an integration time of 1 min or that stated by the manufacturer, whichever is shorter. Table 5 and Table 6 provide guidance regarding identification performance and list the daughters and possible impurities, respectively. Following Table 5, identification results shall be reported as either a) complete and correct, b) incomplete, c) incorrect or, d) incomplete and incorrect. The medical radionuclides (^{99m}Tc, ^{131}I, and ^{201}Tl) shall be surrounded by 8 cm of polymethyl methacrylate (PMMA) for test purposes to represent in-vivo conditions.	6.6.3
Identification of mixed radioactive materials	The following combined radionuclides shall be identified within 1 min or as specified by the manufacturer, whichever is shorter. Each radionuclide shall produce an ambient dose equivalent rate (unshielded) at the reference point of 0,5 µSv h^{-1} (± 30 %) above background: ^{131}I + WGPu, NORM (or surrogate), NORM + HEU, NORM + WGPu	6.6.4
Ambient temperature	RIDs shall function correctly including performing identifications, responding to neutron sources (when applicable), and maintaining an acceptable indication of ambient dose equivalent rate over the temperature range specified in IEC 62706 for hand-held devices (–20 °C to +50 °C).	7.2
Temperature shock	RIDs shall function correctly including performing identifications, responding to neutron sources (when applicable), and maintaining an acceptable indication of ambient dose equivalent rate within 1 h of exposure to rapid temperature changes from 20 °C to –20 °C, –20 °C to 20 °C, 20 °C to 50 °C, and 50 °C to 20 °C with each change being made in less than 5 min.	7.3
Relative humidity	RIDs shall function correctly including performing identifications, responding to neutron sources (when applicable), and maintaining an acceptable indication of ambient dose equivalent rate over the range of relative humidity specified in IEC 62706 for hand-held devices (40 % to 93 % at 35 °C).	7.4
Low/high temperature start up	RIDs shall be able to operate when switched on at the cold temperature limit (−20 ºC) or the high temperature limit (50 ºC).	7.5
Moisture and dust protection	RIDs shall meet the IP classification requirements stated in IEC 62706 for hand-held devices (IP 53).	7.6
Vibration	RIDs shall withstand exposure to vibrations associated with the operation of hand-held or hand-held equipment as defined in IEC 62706, vibration test, hand-held, body worn, portable, and transportable requirements of 0,01 $g^2 \cdot Hz^{-1}$ (spectral density) using 5 Hz and 500 Hz as the frequency endpoints for a period of 1 h.	8.2
Mechanical shock	RIDs shall withstand exposure to shock pulses in accordance with the mechanical shock requirements defined in IEC 62706 of 50 g peak acceleration for a nominal 11 ms in each of three mutually orthogonal axes.	8.3
Impact (microphonics)	RIDs shall be unaffected by microphonic conditions such as those that may occur from low intensity impacts from sharp contact with hard surfaces in accordance with the microphonic/impact requirements for hand-held and body worn devices defined in IEC 62706.	8.4
Electrostatic discharge	RIDs shall not be affected by exposure to electrostatic discharges at intensities of up to 6 kV using the contact discharge technique in accordance with the electrostatic requirements defined in IEC 62706.	9.2
Radiofrequency	RIDs should not be affected by RF fields in accordance with the radiofrequency immunity requirements for all other instrument types as defined in IEC 62706 with the following changes: Frequency range from 80 MHz to 2 500 MHz. Field intensities are 10 $V \cdot m^{-1}$ from 80 MHz to 1 000 MHz and 3 $V \cdot m^{-1}$ from 1 000 MHz to 2 500 MHz.	9.3
Radiated RF emissions	The electromagnetic fields emitted by the RID at 3 m shall be less than what is shown in Table 7 in accordance with IEC 62706.	9.4
Conducted disturbances	RIDs shall not be affected by RF fields that can be conducted onto the instrument through an external conducting cable in accordance with IEC 62706, immunity from conducted RF. RIDs that do not have at least one external conducting cable are excluded.	9.5
Magnetic fields	The RID shall be fully functional when exposed to a 100 $A \cdot m^{-1}$ 50 Hz or 60 Hz magnetic field in accordance with IEC 62706.	9.6

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